William P. Dwyer

Biology Ph.D. Candidate | Knight-Hennessy Scholar

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Research Projects

DINNENY LAB | Ph.D. Candidate
2023 – Present | Stanford University, Department of Biology

- Developing biosensors for correlative light and electron microscopy (CLEM), collab with the Dahlberg Lab at the Stanford Linear Accelerator Center (SLAC). **Rui, Zaoralova, Rui et al. Cell (PMID: pending)**
- Linking moisture perception to development in plant roots. **Dwyer, Torres-Martinez and Dinneny (2025) BMC Bio PMID: 41546060**

RHEE LAB | Postbac Researcher
2020 – 2023 | Carnegie Institution for Science, Department of Plant Biology

- Sorghum Metabolic Atlas: a high-throughput pipeline to identify subcellular localization for hundreds of S. bicolor enzymes. **Karia, Dwyer et al. In prep**
- Arabidopsis thaliana RHAMNOSE 1 condensate formation drives UDP-rhamnose synthesis. **Field, Dorone, Dwyer et al. In prep**
- Plant Metabolic Network 15: A resource of genome-wide metabolism databases for 126 plants and algae. **Hawkins et al. including Dwyer (2021) JIPB PMID: 34403192**

HOUTEN LAB | Summer Research Fellow
2019 | Icahn School of Medicine at Mt. Sinai, Department of Genetics and Genomics

- Acyl-CoA dehydrogenase substrate promiscuity: Challenges and opportunities for development of substrate reduction therapy in disorders of valine and isoleucine metabolism. **Houten, Dodatko, Dwyer, et al. (2023) JIMD PMID: 37309295**

VASSAR COLLEGE | Undergraduate Researcher
2017-2020 | Department of Chemistry, Biochemistry Program

- Spodek-Keimowitz Lab: Investigating heavy metal contamination in Catskill lake sediment using ICP-MS. **Presented at ACS local chapter conference**
- Kennell Lab: honors thesis characterizing a phosphoglycolate phosphatase (PGP) ortholog in Drosophila melanogaster.

Outreach and Advocacy

GROW AND TELL SEMINAR SERIES
2025 - | Stanford University

- Organizer and co-chair of Stanford's plant biology seminar series

UNDOING THE IVORY TOWER
2023 - 2024 | The Good Scientists

- Founder and lead editor of Undoing the Ivory Tower, a monthly newsletter with 350 members covering community-participatory research

INDIGENOUS CROPS RESEARCH
2022 | Rhee Lab

- Advocating for ethical research of indigenous crops, previously 'orphan crops' in a peer-reviewed article and virtual symposia **Dwyer et al. (2022) TIPS PMID: 36163314**

Education

STANFORD UNIVERSITY
Ph.D. Biology
2023 - Present

VASSAR COLLEGE
B.A. Biochemistry
2016 - 2020 | Poughkeepsie, NY
Cum. GPA: 3.80 / 4.0, Dept Honors

Honors

CMB TRAINING GRANT
2024 Cohort
National Institute of General Medical Sciences (NIGMS), NIH

KNIGHT-HENNESSY SCHOLARSHIP
2023 Cohort
Stanford University

CHURCHILL FELLOWSHIP
2020 Nominee and Finalist
Accepted at Cambridge University, Dept of Plant Sciences

SIGMA XI HONOR SOCIETY
2020 Inductee
Vassar College

Languages

French (native)
English (fluent)
Spanish (conversational)

References

José Dinneny, Professor of Biology, HHMI Investigator, Stanford University
dinneny@stanford.edu

Peter Dahlberg, Asst Prof of Photon Science and Structural Biology, Stanford University
pdahlb@stanford.edu

Sue Rhee, Director, Plant Resilience Institute, Michigan State University
rheeseu6@msu.edu